

BANK LOAN REPORT

Data Analytics Dashboard Project

SQL Server • Power BI • Advanced Excel • DAX • Finance Domain

3 Interactive Dashboards | **38,576 Loans** | **\$435.8M Funded** | **\$473M Received**

SQL

Power BI

Advanced Excel

DAX

Finance

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PROJECT OVERVIEW

OBJECTIVE

Build a comprehensive Bank Loan Report to monitor lending activities, track portfolio health, differentiate Good vs Bad loans, and enable data-driven decisions using SQL Server, Power BI, and Advanced Excel.

Domain	Tools Used	Dashboards	Dataset
Banking & Financial Services	SQL · Power BI Advanced Excel	3 Interactive Dashboards Built	38,576 Loans (financial_loan)

WHAT WAS BUILT

- Dashboard 1 – Summary: 5 KPIs, Good/Bad Loan breakdown, Loan Status Grid View
- Dashboard 2 – Overview: 6 chart types (Line, Filled Map, Donut, Bar ×2, TreeMap)
- Dashboard 3 – Details: Consolidated holistic view of all loan records and metrics
- Advanced Excel: PivotTables, COUNTIFS/SUMIFS, conditional formatting, dynamic charts
- SQL: 20+ validated queries for KPIs, MTD, PMTD, MoM, and filtered analysis

TOOLS & TECHNOLOGY STACK

Microsoft SQL Server

Data Storage & Query Validation

20+ queries covering COUNT, SUM, AVG, CASE WHEN, GROUP BY. Validated all KPIs — Total Applications, Funded Amount, MTD, PMTD, MoM, Good/Bad Loan %, Loan Status grid.

Power BI Desktop + DAX

Interactive Dashboard Development

3 dashboards with slicers (Year, Quarter, Month, Grade, Status, Home Ownership). DAX measures for MTD, MoM%, Good/Bad classification, Average Interest Rate, and DTI.

Advanced Excel

Data Analysis & Reporting

PivotTables for cross-tab analysis, SUMIFS/COUNTIFS for conditional aggregations, dynamic charts (line, donut, bar), conditional formatting for loan status, and data validation.

Data Modeling & Domain Knowledge

Finance Domain + Schema Design

Applied banking concepts: DTI ratio, credit scoring, loan lifecycle (Application → Disbursement → Repayment). Designed star schema and optimized table relationships.

POWER BI DASHBOARD — LIVE SCREENSHOT



BANK LOAN DASHBOARD

Overview of Loan Performance & Key Insights



1/1/2021

12/31/2021



All



FILTERS

Year

All

Quarter

All

Month Name

All

emp_length

All

Loan Grade

All

Loan_Status

All

Home_Ownership

All

Total Loan Application



39K

MTD ▲ 11.2% | MoM ▲ 6.9%

Total Funded Amount



435.8M

MTD ▲ 12.4% | MoM ▲ 13.0%

Total Received Amount



473M

MTD ▲ 12.3% | MoM ▲ 15.8%

Average Interest Rate



12.05%

MTD ▲ 102.5% | MoM ▲ 3.5%



Average DTI

13.33%

MTD ▲ 102.5% | MoM ▲ 2.7%

Total Loan Application By Month



Total Loan Application By Term



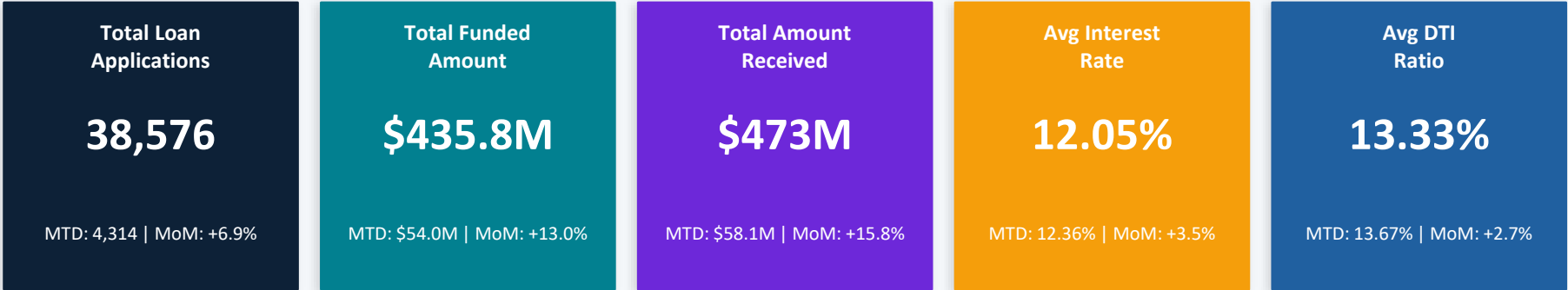
Total Loan Application By Term



Particular	Bad Loan	Current Loan	Good Loan
Loan_Application	5333	1098	32145
Total_Funded_amount	65532225	18866500	351358350
Total Received Amount	37284763	24199914	411586256

emp_length	Loan_Application	Total_Funded_amount	Total Received Amount	Average Interest	Average DTI
10+ years	8870	116115950	125871616	12.09%	13.72%
2 years	4382	44967975	49206961	12.07%	13.10%
< 1 year	4575	44210625	47545011	11.92%	12.55%
3 years	4088	43937850	47551832	12.02%	13.16%
4 years	3428	37600375	40964850	12.17%	13.20%
5 years	3273	36973625	40397571	12.03%	13.43%
Total	38576	435757075	473070933	12.05%	13.33%

DASHBOARD 1: SUMMARY — KPIs & LOAN HEALTH



GOOD LOAN vs BAD LOAN PORTFOLIO ANALYSIS (Real Data from 38,576 Loans)

✓ GOOD LOANS — Fully Paid + Current		✗ BAD LOANS — Charged Off	
Good Loan %:	86.18%	Bad Loan %:	13.82%
Applications:	33,243	Applications:	5,333
Funded Amount:	\$370.2M	Funded Amount:	\$65.5M
Amount Received:	\$435.8M	Amount Received:	\$37.3M

ADVANCED EXCEL — ANALYSIS & TECHNIQUES



PivotTables

- Loan Status breakdown: Bad / Current / Good by applications, funded, received
- Employment length analysis — 10+ years leads with 8,870 applications
- State-wise aggregation: CA (6,894), NY (3,701), FL (2,773) top states
- Cross-tab: Grade × Loan Status to identify risk concentration



SUMIFS & COUNTIFS

- COUNTIFS for MTD: Month = Dec → 4,314 applications
- SUMIFS for Good Loan funded: Status IN {Fully Paid, Current} → \$370.2M
- SUMIFS for Bad Loan received: Status = Charged Off → \$37.3M
- Conditional aggregation by Grade (A–G) and Purpose simultaneously



Dynamic Charts

- Line chart: Monthly trend Jan→Dec showing steady growth (2,332 → 4,314)
- Donut chart: 36-month (73.2%) vs 60-month (26.8%) term split
- Bar chart: Grade distribution — B highest (11,674), G lowest (313)
- Conditional formatting: Red for Charged Off, Green for Fully Paid rows



Excel Formulas Used

- IFERROR wrapping for division-safe interest rate calculations
- TEXT & DATE functions to extract Month Name and Month Number
- AVERAGEIFS for interest rate/DTI by loan grade or employment length
- Data Validation dropdowns for interactive filtering by status/grade

DASHBOARD 2: OVERVIEW — 6 VISUALIZATIONS

Monthly Trends

Line Chart

Jan 2,332 → Dec 4,314

Consistent month-over-month growth throughout 2021, with a notable acceleration in Q4 (Oct–Dec)

Regional Map

Filled Map

CA: 6,894 | NY: 3,701 | FL: 2,773

California dominates with 6,894 loans; top 3 states represent ~34% of total applications

Loan Term Split

Donut Chart

36M: 28,237 (73.2%) | 60M: 10,339 (26.8%)

73.2% borrowers prefer 36-month terms — lower total interest cost drives preference

Employment Length

Bar Chart

10+ yrs: 8,870 | <1yr: 4,575 | 2yrs: 4,382

Experienced employees (10+ years) are the biggest borrowers — stable income = higher credit demand

Loan Purpose

Bar Chart

Debt Consolidation: 18,214 (47.2%)

Debt consolidation accounts for nearly half of all loans, followed by credit cards (13%) and other (9.9%)

Home Ownership

Tree Map

RENT: 18,439 | MORTGAGE: 17,198 | OWN: 2,838

Renters and mortgage holders dominate — both indicate financial need for structured borrowing

SQL QUERIES — DATA VALIDATION LAYER

Every Power BI measure and Excel formula was first validated in SQL Server — ensuring 100% data accuracy across all dashboards

Total Loan Applications → 38,576

```
SELECT COUNT(id) AS Total_Applications
FROM financial_loan
```

MTD Funded Amount (Dec) → \$54M

```
SELECT SUM(loan_amount) AS MTD_Funded
FROM financial_loan
WHERE MONTH(issue_date) = 12
```

Good Loan % → 86.18%

```
SELECT (COUNT(CASE WHEN loan_status IN
('Fully Paid','Current') THEN id END)
*100.0)/COUNT(id) AS Good_Loan_Pct
FROM financial_loan
```

Avg DTI MTD → 13.67%

```
SELECT AVG(dti)*100 AS MTD_Avg_DTI
FROM financial_loan
WHERE MONTH(issue_date) = 12
```

Loan Status Grid View

```
SELECT loan_status,COUNT(id),
SUM(total_payment),SUM(loan_amount),
AVG(int_rate*100),AVG(dti*100)
FROM financial_loan GROUP BY loan_status
```

Grade A Filter (Purpose Breakdown)

```
SELECT purpose,COUNT(id),
SUM(loan_amount),SUM(total_payment)
FROM financial_loan WHERE grade='A'
GROUP BY purpose ORDER BY purpose
```

KEY INSIGHTS FROM REAL DATA



Consistent Growth

Loan applications grew from 2,332 in Jan to 4,314 in Dec — an 85% increase year-over-year, with Q4 being the strongest quarter.



Strong Portfolio Quality

86.18% of 38,576 loans are Good (Fully Paid + Current). Good loans recovered \$435.8M vs \$370.2M funded — positive return.



Charge-Off Risk Alert

5,333 Charged Off loans = \$65.5M funded but only \$37.3M recovered — a \$28.2M loss. DTI monitoring can help reduce defaults.



Debt Consolidation #1

18,214 loans (47.2%) are for debt consolidation — showing borrowers primarily use bank loans to restructure existing financial obligations.



CA Dominates

California alone accounts for 6,894 loans (17.9% of total), followed by NY (9.6%) and FL (7.2%). Strong urban concentration.



Grade B is Highest

Grade B has the most loans (11,674) followed by A (9,689) and C (7,904). Grade G is the riskiest with only 313 applications.

RESUME SUMMARY & KEY POINTS

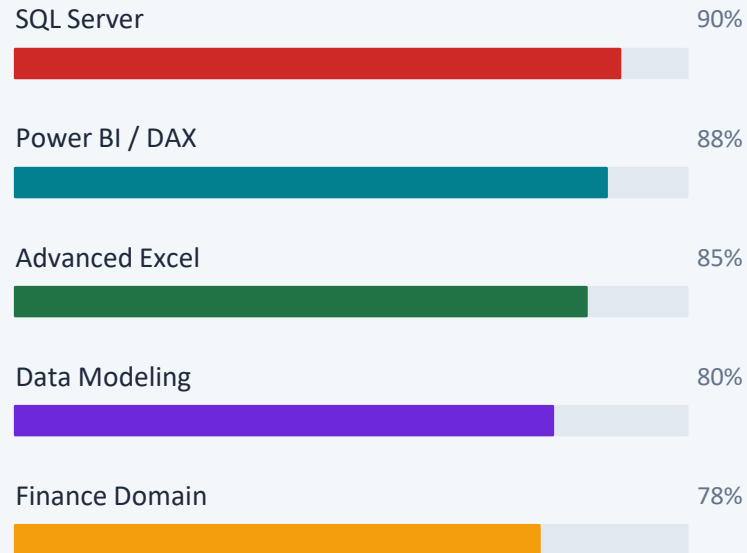
RESUME ONE-LINER:

Built end-to-end Bank Loan Analytics solution using SQL Server, Power BI & Advanced Excel — analysing 38,576 loans, \$435.8M funded, across 3 interactive dashboards with Good/Bad Loan classification and MTD/MoM tracking.

RESUME BULLET POINTS

- Developed 3 Power BI dashboards (Summary, Overview, Details) to monitor \$435.8M loan portfolio with dynamic slicers
- Wrote 20+ SQL Server queries to validate KPIs — COUNT, SUM, AVG, CASE WHEN, MTD & PMTD filtering, MoM% change
- Classified 38,576 loans into Good (86.18%) and Bad (13.82%) using DAX calculated measures and CASE WHEN logic
- Built Advanced Excel models with PivotTables, SUMIFS/COUNTIFS, dynamic charts, and conditional formatting
- Visualised trends using Line, Filled Map, Donut, Bar, and TreeMap charts with cross-filter interactivity
- Applied banking domain knowledge: DTI, credit scoring, loan lifecycle, and portfolio risk classification

SKILLS DEMONSTRATED



CONNECT & EXPLORE THE PROJECT

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LinkedIn Profile

linkedin.com/in/abhinesh-kumar-jaiswal

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